Multivariate Analysis – Honours
Section 1 Exercises Solutions
Chapter 4

Homework exercise 4.1

$$\boldsymbol{X} = \begin{bmatrix} 3 & 6 \\ 4 & 4 \\ 5 & 7 \\ 4 & 7 \end{bmatrix}$$

$$\hat{\boldsymbol{\mu}} = \bar{\boldsymbol{x}} = [4\ 6]'$$

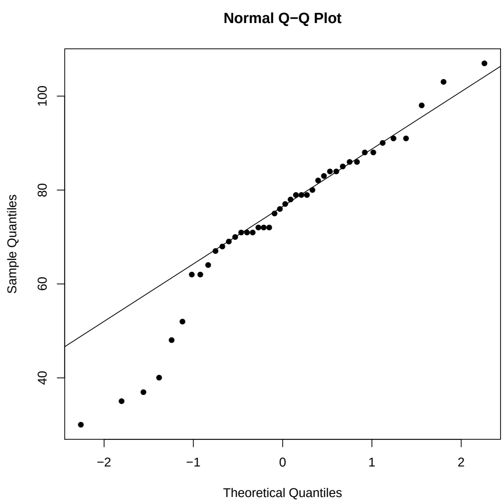
$$\hat{\boldsymbol{\Sigma}} = \frac{1}{n}\boldsymbol{A} = \frac{1}{4}\sum_{j=1}^{4}(\boldsymbol{x}_j - \bar{\boldsymbol{x}}_j)(\boldsymbol{x}_j - \bar{\boldsymbol{x}}_j)' = \frac{1}{4}\begin{bmatrix} 2 & 1 \\ 1 & 6 \end{bmatrix}$$

Homework exercise 4.2

1. $\bar{\boldsymbol{X}} \sim N_6\left(\boldsymbol{\mu}, \frac{1}{60}\boldsymbol{\Sigma}\right)$ and $\sqrt{n}(\bar{\boldsymbol{X}} - \boldsymbol{\mu}) \sim N_6(0, \boldsymbol{\Sigma})$
2. $(\boldsymbol{X}_1 - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{X}_1 - \boldsymbol{\mu}) \sim \chi^2_6$
3. $(n-1)\boldsymbol{S} \sim W_6(59, \boldsymbol{\Sigma})$
4. $n(\bar{\boldsymbol{X}} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\bar{\boldsymbol{X}} - \boldsymbol{\mu}) \sim \chi^2_6$
5. $n(\bar{\boldsymbol{X}} - \boldsymbol{\mu})'\boldsymbol{S}^{-1}(\bar{\boldsymbol{X}} - \boldsymbol{\mu}) \dot{\sim} \chi^2_6$

Homework exercise 4.3

1. Looking at the QQ-plot for the Solar Radiation variable, we see some deviation from normality in the lower tail, although most of the quantiles seem to be in line with that of the normal distribution.



We can calculate r_q as the correlation coefficient between the empirical and theoretical quantiles.

```
airpol <- read.csv('Air Pollution Data.csv')
sol <- airpol$Radiation
qqp <- qqnorm(sol, pch = 16)
round(cor(qqp$x, qqp$y), 4)
```

```
[1] 0.9693
```

Our sample size is $n = 42$. Therefore, to compare this value to the critical values in Table B.1, we need to interpolate the critical values corresponding to $n = 40$ and $n = 45$, yielding the following:

Sample Size (n)	$\alpha = 0.01$	$\alpha = 0.05$	$\alpha = 0.1$
42	0.9612	0.9735	0.9779

Since our test statistic $r_q = 0.9693 > 0.9612$, we will not reject the hypothesis of normality at a significance level of $\alpha = 0.01$. However, we would have rejected at a significance level of $\alpha = 0.05$, since $r_q = 0.9693 < 0.9735$.

From this we can conclude that $0.01 < p - value < 0.05$ (probably closer to 0.05).

Therefore, our conclusion is that there is substantial evidence to suggest that the solar radiation variable is NOT normally distributed, and one should proceed with caution in further analysis which may assume normality.

2. Calculating the squared distances:

```
(d2 <- mahalanobis(biv, colMeans(biv), cov = var(biv)))
```

```
 [1]  0.4606524  0.6592206  2.3770610  1.6282902  0.4135364  0.4760726
 [7]  1.1848895 10.6391792  0.1388339  0.8162468  1.3566301  0.6228096
[13]  5.6494392  0.3159498  0.4135364  0.1224973  0.8987982  4.7646873
[19]  3.0089122  0.6592206  2.7741416  1.0360061  0.7874152  3.4437748
[25]  6.1488606  1.0360061  0.1388339  0.8856041  0.1379719  2.2488867
[31]  0.1901188  0.4606524  1.1471939  7.0857237  1.4584229  0.1224973
[37]  1.8984708  2.7782596  8.4730649  0.6370218  0.7032485  1.8013611
```

3. If these variables were bivariate normally distributed, one would expect approximately 50% of the squared distances above to be below $\chi^2_2(0.5) = 1.386$.

```
mean(d2 < qchisq(0.5, 2))
```

```
[1] 0.6190476
```

Approximately 62% of the distances are below the 50% quantile. This indicates deviation from normality.

4. The chi-square plot also suggests some deviation from bivariate normality: